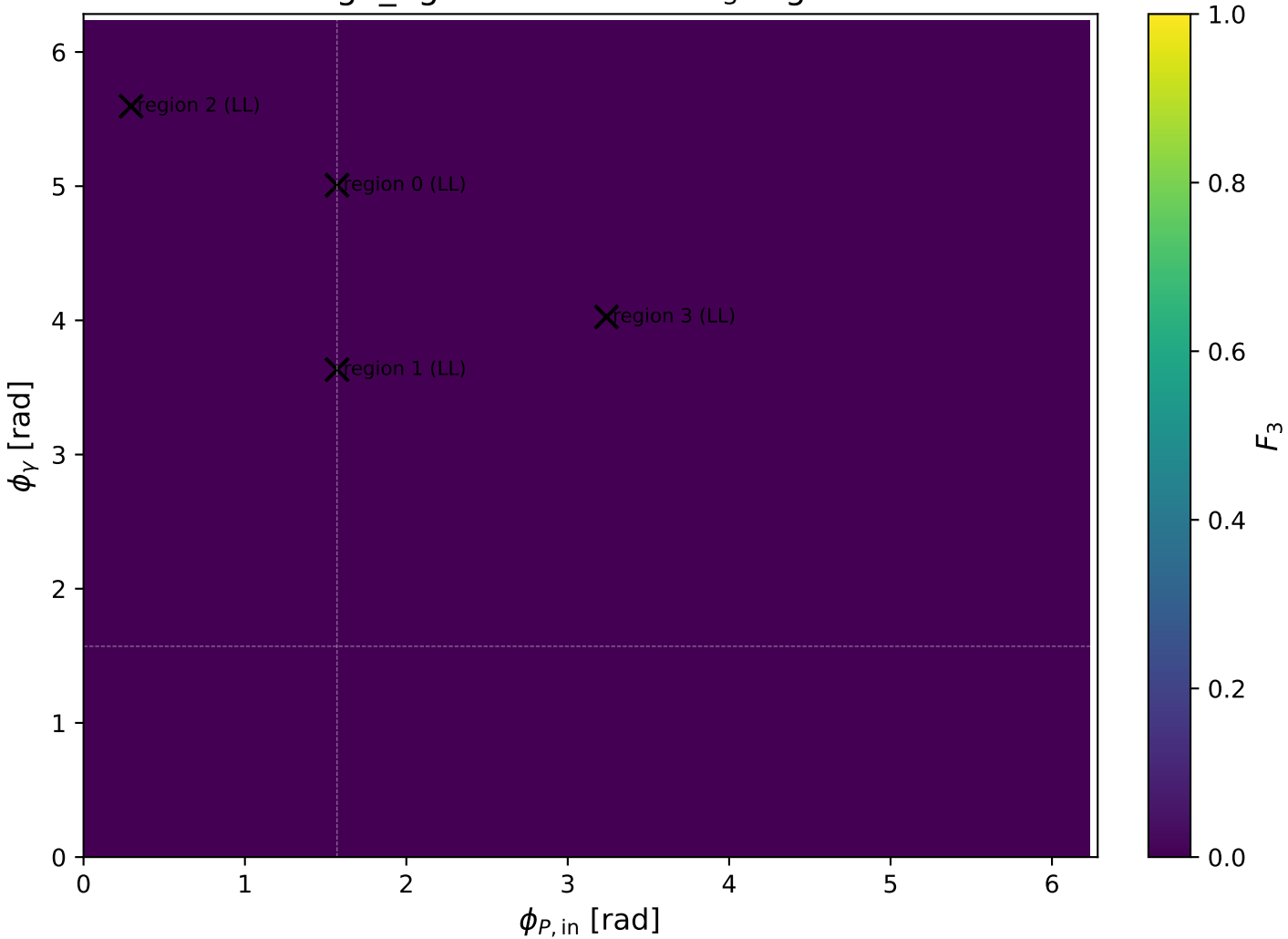
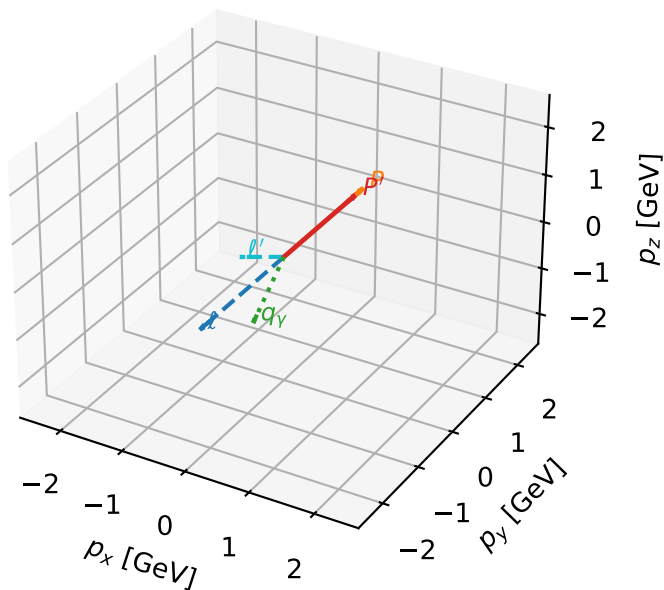


high\_Egamma LL: max  $F_3$  regions



# LL characteristic max $F_3$ configuration

3D momenta

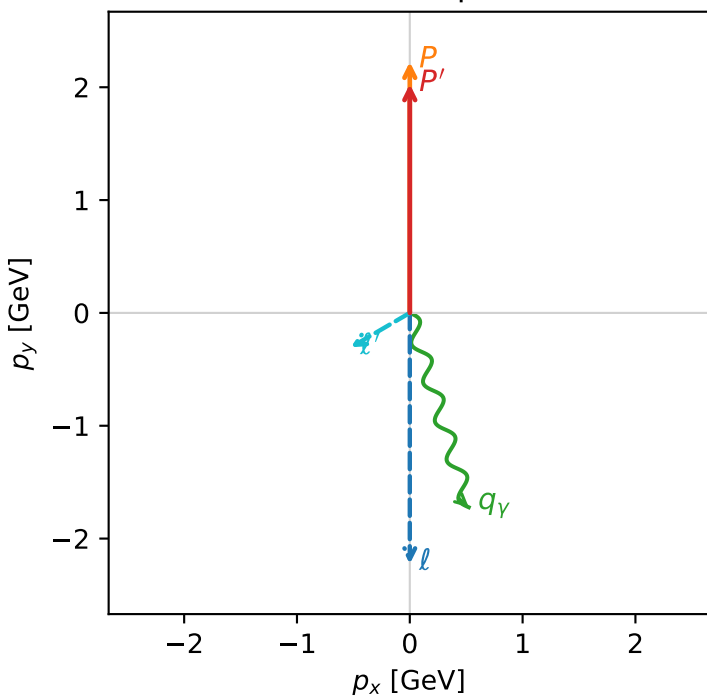


f3\_high\_egamma\_ll\_region\_0 F\_3=9.73022e-09  
 region: high\_Egamma\_LL\_region\_0  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
 incoming state:  $|L_e L_p\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.02717$   
 $\theta_{in}=1.57079$ ,  $\phi_l=4.71239$ ,  $\phi_p=1.5708$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=5.00691$   
 $Q^2=1.33544$ ,  $x_B=0.0816271$ ,  $t=-0.00635022$ ,  $W^2=15.9046$ ,  $y=0.792799$   
 $\theta(l', \gamma)=76.6286$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2244$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $l'$   $E= 0.6049$   $p_x= -0.5225$   $p_y= -0.3047$   $p_z= 0.0000$   
 $P'$   $E= 2.2337$   $p_x= 0.0000$   $p_y= 2.0272$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.8000$   $p_x= 0.5225$   $p_y= -1.7225$   $p_z= 0.0000$

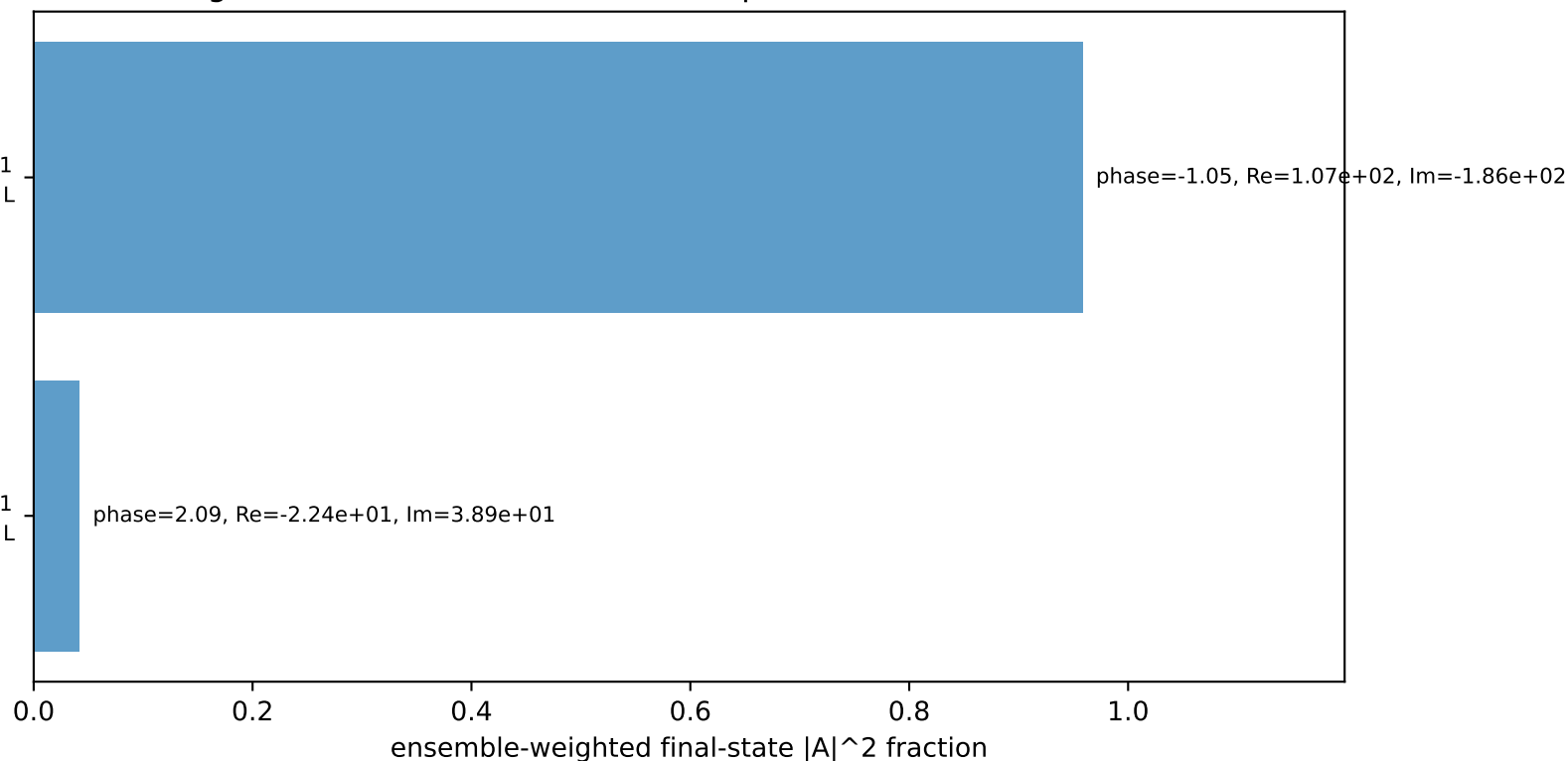
Transverse plane



$h_e=+1, h_p=-1, h_\gamma=+1$   
 electron L, proton L

$h_e=+1, h_p=+1, h_\gamma=-1$   
 electron L, proton L

Leading initial-ensemble/final-state components (N=2, retained=100.0%)



# LL characteristic max $F_3$ configuration

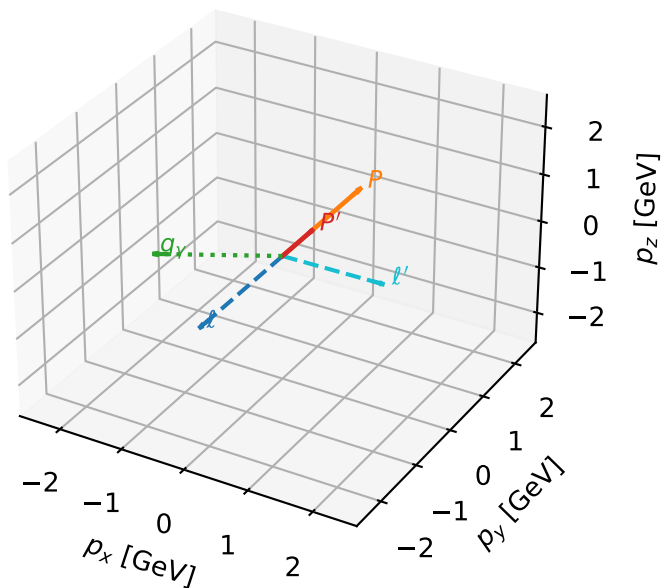
f3\_high\_egamma\_ll\_region\_1 F\_3=4.15877e-09  
region: high\_Egamma\_LL\_region\_1  
outgoing-state purity: 1  
kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
incoming state:  $|L_e L_p\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=0.827624$   
 $\theta_{in}=1.57079$ ,  $\phi_l=4.71239$ ,  $\phi_p=1.5708$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=3.63247$   
 $Q^2=7.15589$ ,  $x_B=0.547769$ ,  $t=-0.59805$ ,  $W^2=6.78766$ ,  $y=0.633054$   
 $\theta(l', \gamma)=152.629$  deg

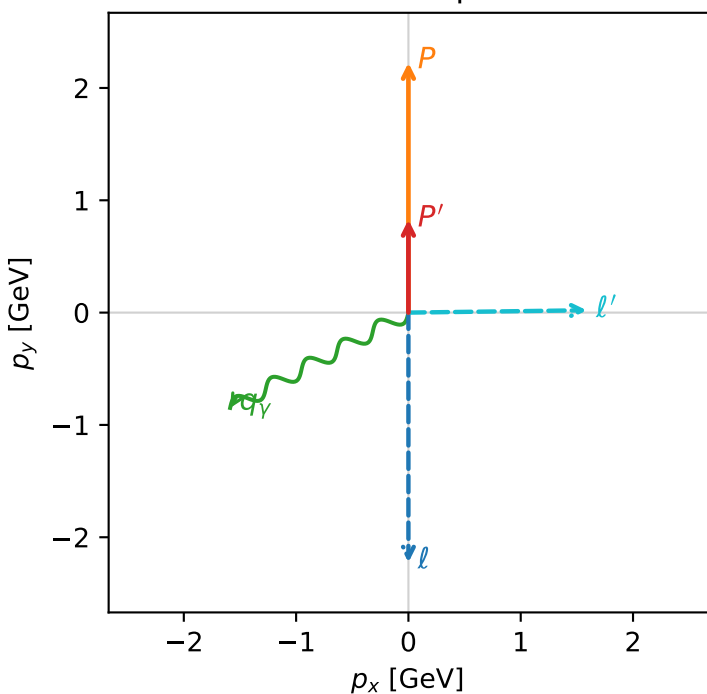
four-momenta  $[E, p_x, p_y, p_z]$  GeV:

|            |      |        |        |         |        |         |        |         |
|------------|------|--------|--------|---------|--------|---------|--------|---------|
| $l$        | $E=$ | 2.2244 | $p_x=$ | -0.0000 | $p_y=$ | -2.2244 | $p_z=$ | -0.0000 |
| $P$        | $E=$ | 2.4141 | $p_x=$ | 0.0000  | $p_y=$ | 2.2244  | $p_z=$ | 0.0000  |
| $l'$       | $E=$ | 1.5876 | $p_x=$ | 1.5875  | $p_y=$ | 0.0209  | $p_z=$ | 0.0000  |
| $P'$       | $E=$ | 1.2509 | $p_x=$ | 0.0000  | $p_y=$ | 0.8276  | $p_z=$ | 0.0000  |
| $q_\gamma$ | $E=$ | 1.8000 | $p_x=$ | -1.5875 | $p_y=$ | -0.8485 | $p_z=$ | 0.0000  |

3D momenta

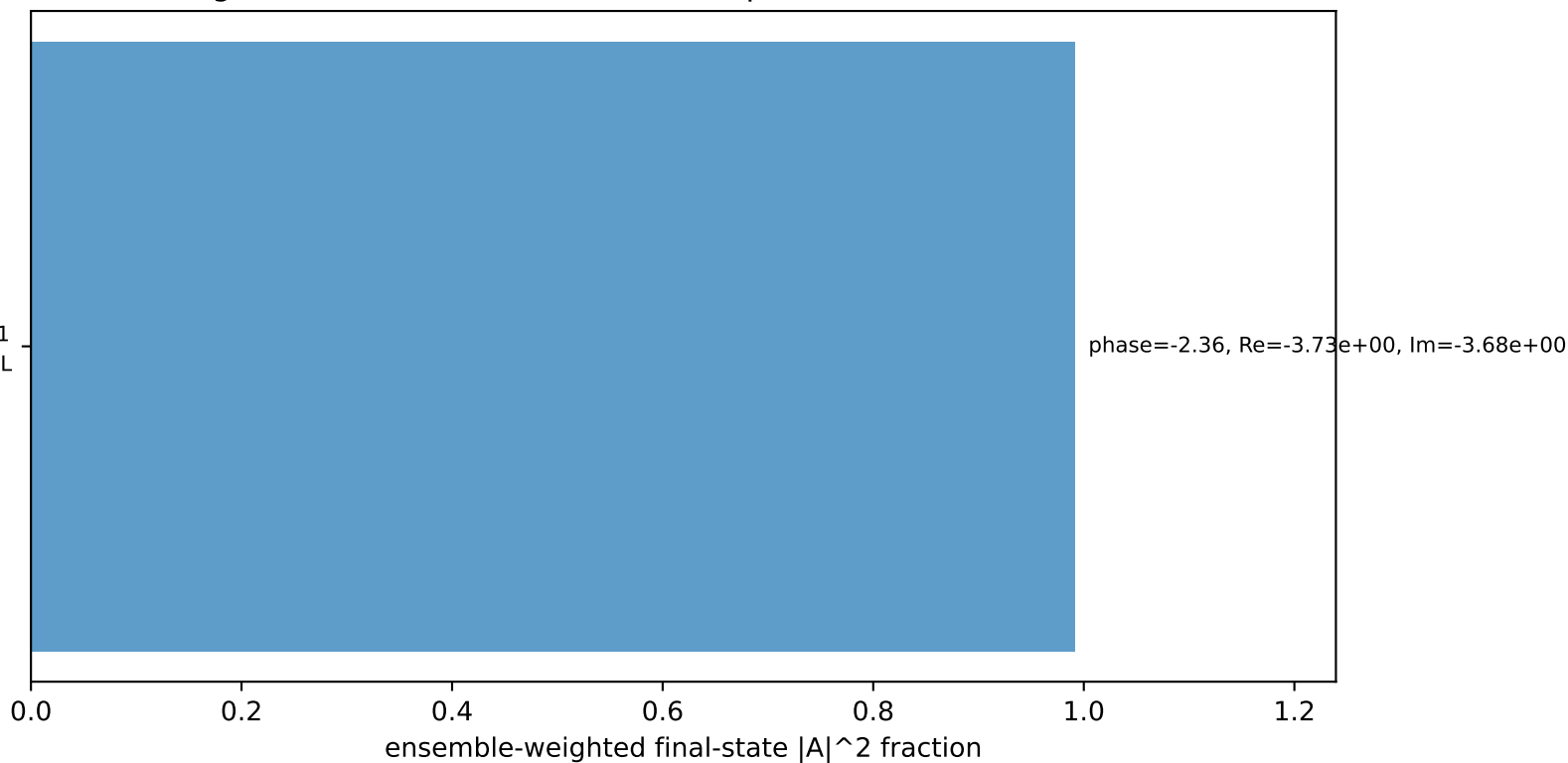


Transverse plane



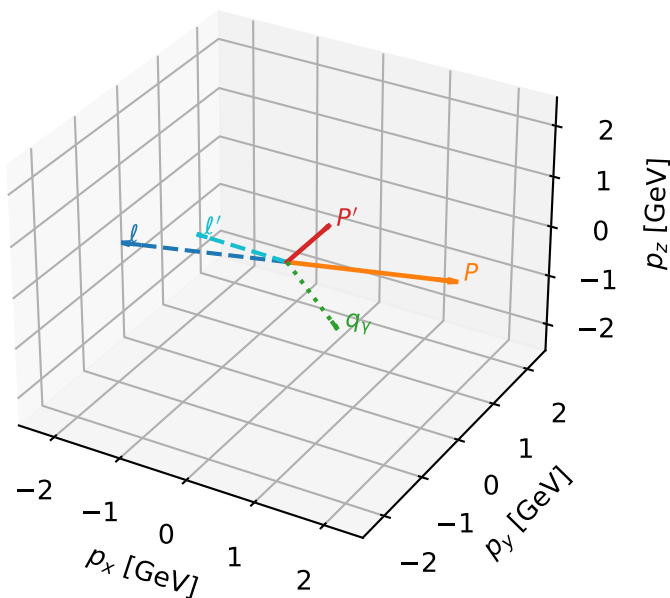
$h_e=+1$ ,  $h_p=-1$ ,  $h_\gamma=+1$   
electron L, proton L

Leading initial-ensemble/final-state components (N=1, retained=99.2%)



# LL characteristic max $F_3$ configuration

3D momenta

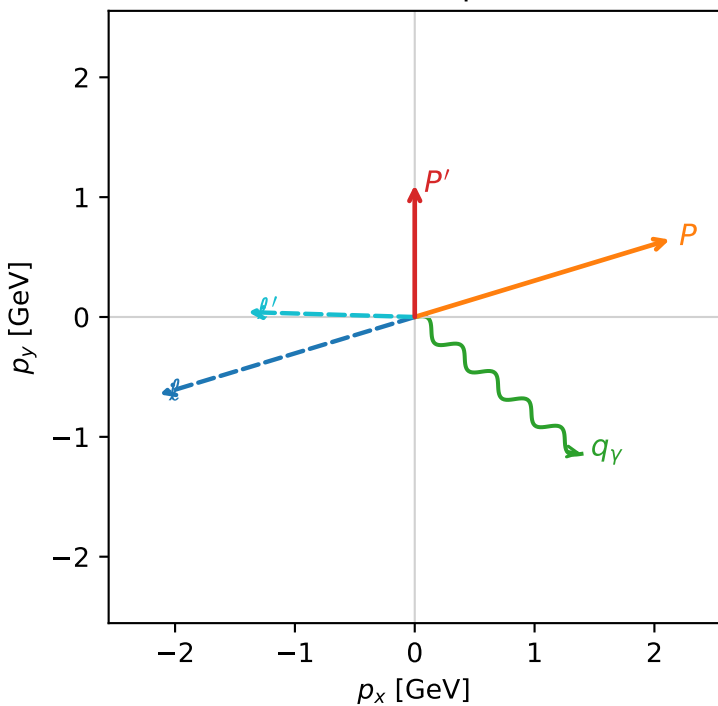


f3\_high\_egamma\_ll\_region\_2 F\_3=3.69056e-09  
region: high\_Egamma\_LL\_region\_2  
outgoing-state purity: 1  
kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
incoming state:  $|L_e L_p\rangle$

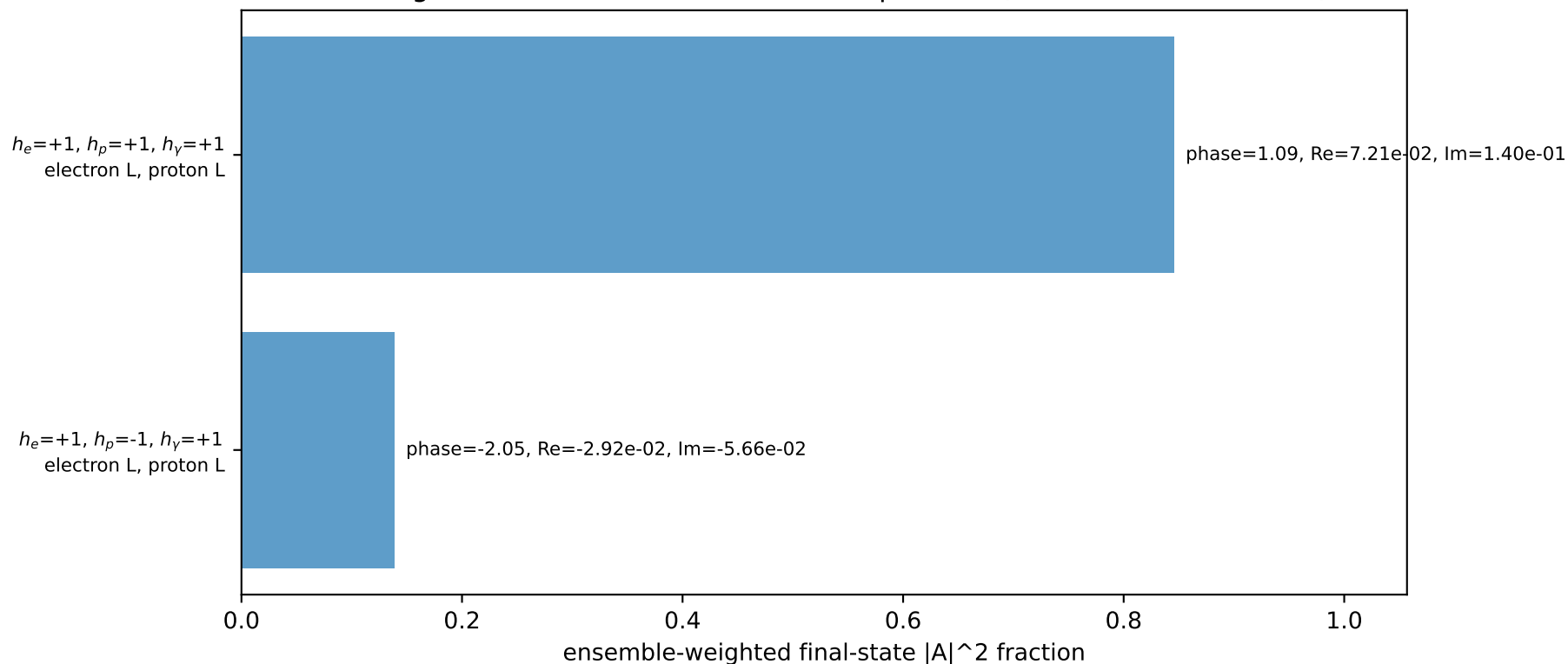
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=1.10115$   
 $\theta_{in}=1.57079$ ,  $\phi_l=3.43612$ ,  $\phi_p=0.294524$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=5.59596$   
 $Q^2=0.321844$ ,  $x_B=0.0400101$ ,  $t=-3.80226$ ,  $W^2=8.60207$ ,  $y=0.389807$   
 $\theta(l', \gamma)=142.303$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2244$   $p_x= -2.1286$   $p_y= -0.6457$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 2.1286$   $p_y= 0.6457$   $p_z= 0.0000$   
 $l'$   $E= 1.3920$   $p_x= -1.3914$   $p_y= 0.0408$   $p_z= 0.0000$   
 $P'$   $E= 1.4465$   $p_x= 0.0000$   $p_y= 1.1011$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.8000$   $p_x= 1.3914$   $p_y= -1.1419$   $p_z= 0.0000$

Transverse plane

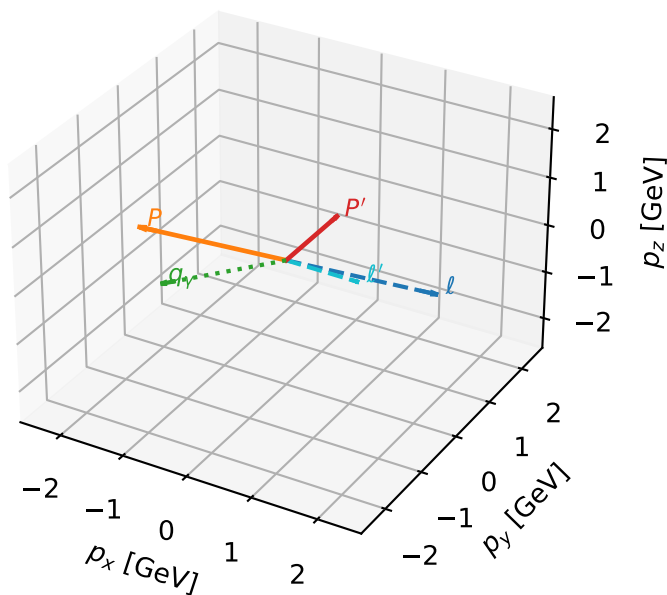


Leading initial-ensemble/final-state components (N=2, retained=98.4%)



# LL characteristic max $F_3$ configuration

3D momenta

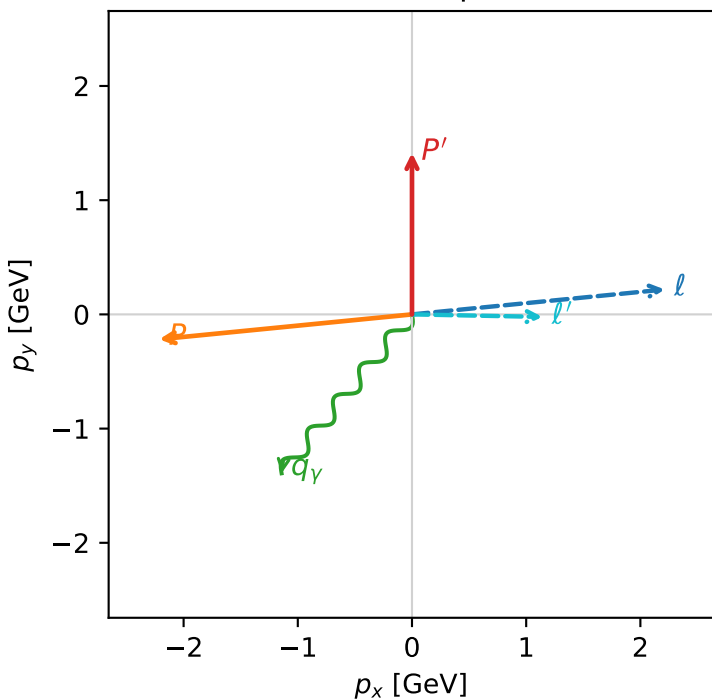


f3\_high\_egamma\_ll\_region\_3 F\_3=3.52706e-09  
 region: high\_Egamma\_LL\_region\_3  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
 incoming state:  $|L_e L_p\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=1.41348$   
 $\theta_{in}=1.57079$ ,  $\phi_l=0.0981748$ ,  $\phi_P=3.23977$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=4.02517$   
 $Q^2=0.0350288$ ,  $x_B=0.00347662$ ,  $t=-7.04722$ ,  $W^2=10.9203$ ,  $y=0.48825$   
 $\theta(l', \gamma)=128.268$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2244$   $p_x= 2.2137$   $p_y= 0.2180$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= -2.2137$   $p_y= -0.2180$   $p_z= 0.0000$   
 $l'$   $E= 1.1421$   $p_x= 1.1419$   $p_y= -0.0221$   $p_z= 0.0000$   
 $P'$   $E= 1.6964$   $p_x= 0.0000$   $p_y= 1.4135$   $p_z= 0.0000$   
 $q_Y$   $E= 1.8000$   $p_x= -1.1419$   $p_y= -1.3914$   $p_z= 0.0000$

Transverse plane



$h_e=+1, h_p=+1, h_\gamma=+1$   
 electron L, proton L

$h_e=+1, h_p=-1, h_\gamma=+1$   
 electron L, proton L

Leading initial-ensemble/final-state components (N=2, retained=97.5%)

